PROFINITE TOPOLOGICAL SPACES HAVE NO HIGHER COHOMOLOGY FOR ANY ABELIAN SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and is three sentences long in the source; nothing is referred to the reader.

Editorial note. The counted spans carry 2 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-cohomology/retained-preamble.tex lines 1-81), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-vanishing-for-profinite is defined only in cohomology.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Every open covering of a profinite space is refined by a finite disjoint union decomposition into open parts. Topology, Lemma topology-lemma-profinite-refine-open-covering, invoked at proof lines 3923-3925. Supplies the refinement that makes the global section functor exact.
- The higher derived functors of an exact functor vanish. Derived Categories, Lemma derived-lemma-right-derived-exact-functor, invoked at proof lines 3929-3930. Supplies the conclusion from exactness of the global section functor.

Source reference. Original lemma 015F: sources/stacks-chow/tags.txt, source lines 1506–1506. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 08ZZ: sources/stacks-chow/tags.txt, source lines 11392–11392. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let X be a profinite topological space. Then $H^q(X, \mathcal{F}) = 0$ for all $q > 0$ and all abelian sheaves $\mathcal{F}$.*

Proof. Any open covering of X can be refined by a finite disjoint union decomposition with open parts, see Topology, Lemma 08ZZ. Hence if $\mathcal{F} \to \mathcal{G}$ is a surjection of abelian sheaves on X, then $\mathcal{F}(X) \to \mathcal{G}(X)$ is surjective. In other words, the global sections functor is an exact functor. Therefore its higher derived functors are zero, see Derived Categories, Lemma 015F. □

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